Title: We have a new way to fight the deadliest animal on earth, why aren't we using it?

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If you had the power to give mosquitoes a new trait, what would you do? What would you change?

If I were asked this question 5 years ago, I might naively have said, "make them unable to bite humans." While it sounds nice, the problem is that that trait would not spread into the mosquito population. This is because female mosquitoes need blood to have eggs and pass their traits on to the next generation.

So what *should* we do? This is a question I have been working on since starting at NC State as a graduate student in biomathematics. The technology that allows the introduction of a new trait into mosquito populations is called a "gene drive". Scientists have been testing what we can do with gene drives for more than a decade. Using gene drives, scientists can introduce many different traits into pest populations. It started by making flies green, but has progressed to much more impactful applications.

In mosquitoes, the focus of gene drives is on disease prevention. Mosquitoes spread malaria, dengue, Zika, and several other deadly diseases. The tiny mosquito spreads so much disease that it is actually the deadliest animal on earth.

In labs, gene drives have shown promise in preventing mosquitoes using two strategies. First, they can be used to spread a trait that prevents mosquitoes from spreading a specific disease. Second, they can crash mosquito populations by spreading a trait that causes mosquitoes to only have male offspring. Either approach, if deployed responsibly, has the potential to save lives.

So why have we not seen any of these gene drives in action? Well, with their power to do good comes their potential to go wrong. One of the central tenets of a gene drive is that they are able to spread quickly into populations, but the problem is that we do not want them to spread too far.

For example, we may be trying to eradicate an invasive species of mosquito that is spreading dengue to a new region. We would not want the gene drive release in the invasive population to spread back to the native population in another country that may not want the gene drive. Scientists like myself have come up with strategies to prevent this, but predicting how well they will work is incredibly challenging without seeing a field test. We have also come up with strategies that can cause a gene drive to break down after a year or so, so that they stop spreading and are only temporary in an area.

With this in mind, we need people like you to call your local representative and push for gene drives to be tested in a controlled field setting on an island, where the potential for them to spread is limited, and they will fallout of the population after a short amount of time. This will allow us to make a more informed decision about whether to use gene drives in a more impactful context. We will be able to collect the data necessary to predict their benefits and risks more effectively. Gene drives have the potential to save many lives, but that potential can never be reached if we never progress with gene drive testing. There is only so much we can do in labs before it becomes necessary to see them in the wild.

Mosquitoes are the deadliest animal on earth, so let's give them a trait that takes that title away from them.